

**Technical Data****BYPASS COOLER**

Drawing Number	-----	3-16680-00345
Type	-----	AEM
Number of coolers	-----	1
Quantity of Hydro-Carbon gas (syngas)	Kg/Hr.	410.5 X 1000
Gas Temperature Inlet / Outlet	Deg.C.	105/95
Pressure drop on gas side	Kg/Sq.Cm	1.6
Quantity of Cooling water	Cu.M/Hr.	225
Water Temperature Inlet/Outlet	Deg.C	35/45
Pressure drop on water side	Kg/Sq.Cmg	0.35
Cooling surface area	Sq.M.	77
Size of tubes: O.D. x Thk - L	mm	19.05X2.4-3000
Velocity of cooling Water / Gas	M/Sec	1.85/11.3
Total number of tubes	---	432
Number of Water / Gas passes	---	2/1
Design Pressure Water / Gas side	Kg/Sq.Cm	6/170
Design Temperature Water / Gas side	Deg.C	60/140
Test pressure Water / Gas side	Kg/Sq.Cm	9/255
Shell side medium	---	SYNGAS
Tube side medium	---	WATER
<u>MATERIALS :</u>		
Shell & Channel	CS - SA 516 GR 70	
Chennel Cover	CS - SA 105	
Tube plates	CS - SA 350 LF 2	
Tubes	CS - SA 210 GR C	
Baffle plates	CS - IS 2062	
Inlet & Otlet Nozzles	CS - SA 105	
Nozzles flanges	CS - SA 105	
Bolts & Nuts	CS - SA 193 B7 / 194 2H	
Gaskets	IRON JACKETED ASBESTOS	
<u>WEIGHTS :</u>		
Gas cooler dry	Kgs	12,000.0
Gas cooler in operation	Kgs	14,000.0
Gas cooler flooded	Kgs	15,500.0